

70. CALCULATING VO2MAX WITHOUT LAB TEST

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VO2max (maximum oxygen consumption) is the maximum amount of oxygen that the body can use per minute to produce energy (watts). So, it is a metric representing the horsepower of the athlete's aerobic engine. It can be expressed as Litres per Minute (L/Min) or milliliters per kilogram per minute (ml/kg/min), with the latter being a better option as normalised for body weight. In fact it is intuitive to understand that a 2.00m tall athlete is likely to be able to use more L/min of oxygen than a 1.60m one.

VO2max is the main factor determining cycling performance: large positive correlations have been reported between VO2max in ml/min/kg and Watts/Kg (or VO2max in L/Min and Watts) for performance lasting approximately between 5 min and 1 hour. This means that, in such efforts, a rider with 80 VO2max will very likely outperform a rider with 65 VO2max, who in turn will very likely outperform a rider with 50 VO2max.

When VO2max differences are small (I would say maximum 5-10 ml/kg/min) other physiological factors can contribute to determine races' outcome:

1. Having Lactate/Ventilatory thresholds/CP/FTP at a high percentage of VO2max to optimize performance in efforts lasting ~20min-1hour.

2. Efficiency: the ability to spend as little energy as possible to produce a certain amount of watts (it depends on physiological and biomechanical factors)
3. Anaerobic Reserve: to launch winning high power efforts lasting ~1s-2min through the anaerobic metabolism.
4. Durability: the ability to maintain performance level during very long races
5. Recovery: the ability to maintain the highest possible performance level in multiday races (one-week stage races and grand tours). Having a great recovery can also help to load more during training and so to make more improvements over time.

So, how to measure VO2Max?

The gold standard to measure VO2max is attending a physiology lab and doing an **incremental maximal exercise test measuring gas expired through a metabolic cart**. The price for such a test is usually between 150 and 400 euros. As you are spending money, you would like to have accurate test results, but this is not always the case. In fact, here are several metabolic cart models on the market, but not all are enough accurate. If you are considering to do a lab test, [here](#) you can find a review about the accuracy of different metabolic cart models to choose the right lab to attend.

However, for money and logistics reasons, you might not attend a lab test to know your VO2max or the VO2max of the athletes you work with.

Can a field test with using only power meter data estimate accurately VO2max?

A study published in 2021 by Sebastian Sitko and Colleagues (Universidad de Zaragoza, Spain) on *International Journal of Sport Physiology and Performance* tried to answer this question. (1)

WHAT DID THEY DO?

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- 46 road cyclists (11 professionals) with VO2max (measured with a metabolic cart) ranging from ~45 to ~80 ml/min/kg were recruited
 - On two different days separated by 72 hrs of rest, they performed:
 1. A VO2max lab test (maximal incremental step test) measuring gas expired through a metabolic cart
 2. A 5 min all-out test

The goal was to see if using some mathematical modelling, the VO2max measured in the lab could be estimated using only the power output recorded in the 5 min all-out test.

Both tests were performed on athletes' own bike mounted on a Tacx indoor trainer.

WHAT DID THEY FIND?

VO₂max was estimated quite accurately from the 5min time trial using the following equation:

$$\text{VO}_2\text{max (ml/min/kg)} = 16.61 + 8.87 \times \text{5min power output (W/kg)}$$

How much accurately? The equation can estimate VO₂max within a 3% error 95% of the times. This means that, for example, if you score 6.8 W/Kg on 5min:

$$\text{VO}_2\text{max (mL/min/kg)} = 16.61 + 8.87 \times 6.8 = 76.9 \text{ ml/min/kg}$$

So, if you have 6.8 W/kg on 5min, there is 95% of probability that your VO₂max is between 74.6 and 79.2 (76.9 ± 2.3 ml/min/kg). 2.3 ml/min/kg is 3% of 76.9 ml/min/kg.

PRACTICAL TAKEAWAYS

VO₂max can be estimated quite accurately using only the 5 minute W/Kg max mean power output.

GABRIELE'S COMMENT

How can we use these results?

- For cyclists who can't attend a physiology lab to perform a test with a metabolic cart, using the above equation could be a valid alternative to have an idea of their VO₂max.
- **Of course, this is an estimation, and a lab test is still the gold standard to know exactly the VO₂max value.**
- To estimate VO₂max with this method. power meter should be accurate and calibrated to not underestimate or overestimate the VO₂max value.

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCES:

1. Sitko S, Cirer-Sastre R, Corbi F, López-Laval I. Five-Minute Power-Based Test to Predict Maximal Oxygen Consumption in Road Cycling. Int J Sports Physiol Perform. 2022 Jan 1;17(1):9-15.





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